

Khwaja Mohammed Junaid Shaik

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SUMMARY

Recent graduate building RAG pipelines and LLM applications with LangChain, LangGraph, and OpenAI/Groq APIs, with a focus on reliability and safe deployment through guardrails, model fallback, FastAPI, MLflow, Docker, and Azure.

SKILLS

Languages: Python, SQL(PostgreSQL)

Machine Learning / Deep Learning: TensorFlow, Keras, Scikit-learn, NLTK

GenAI: LangChain, LangGraph, LangSmith, RAG, Prompt Engineering, OpenAI API, Groq API, Hugging Face

Backend Development: FastAPI, Flask

Data handling & Visualization: Pandas, NumPy, Matplotlib

Developer Tools: Git/GitHub, GitHub Actions, Docker, Mlflow

Cloud: Azure

PROJECTS

Supply Chain Intelligence System – Forecasting, Inventory & Agentic Analytics Platform

<https://github.com/gyrfalcon55/supply-chain-intelligence-system>

- Engineered demand forecasting for 200 SKUs using Nixtla's CrostonClassic for intermittent demand, achieving MAE 2.96, RMSE 3.33, and MASE 0.65.
- Built procurement and inventory decision logic to assess stockout risk, reconcile open orders, and classify inventory as CRITICAL, AT_RISK, or SUFFICIENT.
- Developed a LangChain/LangGraph analytics agent with PostgreSQL MCP integration, adding model fallback and AI guardrails for improved reliability and safety.
- Containerized and deployed the FastAPI, Streamlit, PostgreSQL, and MLflow stack on Azure using Docker Compose and GitHub Actions CI/CD.
- Tech Stack:** Python, FastAPI, Streamlit, PostgreSQL, Nixtla, MLflow, LangChain, LangGraph, MCP, Docker, GitHub Actions, Azure

Skin Lesion Classification and Report Generation using RAG

<https://github.com/gyrfalcon55/Skin-Lesion-Diagnosis-and-Report-Generation>

- Trained an EfficientNet model on the HAM10000 dataset to classify 7 skin lesion types, reaching **89% weighted F1-score**.
- Added Grad-CAM visualizations to show which image regions drove each prediction, paired with LangChain/OpenAI-generated diagnostic summaries.
- Deployed the app on Azure with Streamlit for real-time image upload, prediction, and report generation.
- Tech Stack:** Python, TensorFlow, Keras, OpenCV, Flask, OpenAI API, LangChain

IMDB Sentiment Analysis – Multi-Model NLP Pipeline with MLflow & FastAPI

https://github.com/gyrfalcon55/IMDB_Sentiment_Analysis

- Benchmarked 4 classification models (Logistic Regression, SVM, Random Forest, Naive Bayes) on 50,000 IMDB movie reviews, tuning each with RandomizedSearchCV.
- Reached **87.6% F1-score** and **0.879 ROC-AUC** with the top model (Logistic Regression), logging CV score, F1, and ROC-AUC for every run to MLflow on DagsHub.
- Selected the best model automatically and served predictions through a FastAPI app.
- Tech Stack:** Python, Scikit-learn, NLTK, MLflow, DagsHub, DVC, FastAPI, Docker, Git

EXPERIENCE

AICTE – AI-ML Virtual Internship

Apr 2025 – Jun 2025

- Completed a AI-ML virtual internship backed by Google for Developers under AICTE's India Edu Program, covering AI/ML fundamentals, neural networks, and TensorFlow.
- Built a Loan Eligibility Prediction app using Logistic Regression, GridSearchCV, Flask, and Bootstrap, achieving **89% accuracy**.

EDUCATION

B.Tech Computer Science & Engineering (Data Science)

2022 - 2026

Raghu Engineering College

CGPA: 8.1

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate (link)